Summary Sheet

ProteinVision

RDKit Molecular Descriptors

Quick reference for turning molecular structures into numerical molecular descriptors and preparing them for analysis.

1. What descriptors represent

- A scalar descriptor gives one number for one aspect of a molecule.
- Values may be counts or continuous quantities: The RDKit descriptors library has 217 descriptors (110 count variables, 107 float variables)
- Descriptors can help with molecular comparison, dataset exploration and molecular property prediction.

2. Examples of how descriptors are calculated

- **Molecular mass (daltons, Da):** Add the atomic masses of all atoms, including implicit hydrogens. Use average atomic masses when isotopes are unspecified.
- **Hydrogen bond donor count:** Count eligible sites whose hydrogen can participate in a hydrogen bond. Count donor sites, not all hydrogen atoms.
- **Topological polar surface area (TPSA):** Estimate polar surface area in square angstroms (Å²) by summing predefined contributions from polar atom environments (atom types and bonding). This estimation uses connectivity; no 3D structure is required.
- **Octanol–water partitioning (logP):** Sum contributions from atom types and their bonding environments to estimate the base-10 logarithm of the equilibrium concentration ratio (octanol / water) for a neutral molecule. Higher values mean greater preference for octanol.

The counts are exact for a given structure and a set of counting rules. TPSA and logP are estimates of physical properties, not direct measurements.

3. (Pre-)Processing of molecules and descriptors

- Validate each structure and exclude empty molecules. Identify repeated structures.
- Calculate the same descriptors for every molecule.
- A valid structure can still give an undefined or failed calculation. Treat infinite results as missing; keep valid zeros.
- A descriptor's median is a simple replacement for missing values; but it does not recover the true value.
- You might want to remove constant columns because they cannot help to distinguish molecules.

4. Redundant information

- Some descriptors are highly redundant to other descriptors
- Removing strongly correlated descriptors can simplify the inputs.

Prepared by the ProVis ProteinVision GmbH (www.protein.vision). Last modified on 28.10.2026.

Appendix: Name and Description of all 217 descriptors

Name: MaxAbsEStateIndex
Category: regression
Value range: 0 or greater
Description: Largest absolute atom EState value; reports the strongest electronic-environment score regardless of whether that score is positive or negative.

Name: MaxEStateIndex
Category: regression
Value range: No fixed bounds; signed values
Description: Highest atom electrotopological state (EState) value; combines an atom's electronic character with the influence of other atoms through the bond network.

Name: MinAbsEStateIndex
Category: regression
Value range: 0 or greater
Description: Smallest absolute atom EState value; identifies the atom whose electronic-environment score lies closest to zero, ignoring its sign.

Name: MinEStateIndex
Category: regression
Value range: No fixed bounds; signed values
Description: Lowest atom electrotopological state (EState) value; identifies the lowest signed electronic-environment score among the atoms in the molecule.

Name: qed
Category: regression
Value range: 0 to 1 inclusive
Description: A score from zero to one combining eight molecular properties to estimate similarity to typical oral drugs; higher values indicate greater drug-likeness.

Name: MolLogP
Category: regression
Value range: No fixed bounds; signed values
Description: Wildman-Crippen estimate of the octanol/water partition coefficient on a base-10 logarithmic scale; higher values indicate greater preference for a nonpolar environment.

Name: MolMR
Category: regression
Value range: 0 or greater
Description: Wildman-Crippen estimate of molar refractivity from atom contributions; reflects molecular polarizability and size, describing how the molecule interacts with light.

Name: SPS
Category: regression
Value range: 0 or greater
Description: Spacial score per heavy atom; summarizes structural complexity using atom hybridization, stereochemical features, non-aromatic rings, and branching in the molecular graph.

Name: AvgIpc
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: Average information content of a mathematical description of the bond network; a size-normalized measure of structural information in the hydrogen-suppressed molecular graph.

Name: BalabanJ
Category: regression
Value range: Real number; may be negative, zero or positive

Description: A graph connectivity index using shortest bond-path distances and ring structure; describes how atoms are connected and how compactly the network is arranged.

Name: BertzCT
Category: regression
Value range: Real number; may be negative, zero or positive
Description: A structural-complexity index combining the variety of bonding arrangements and atom types; larger values generally indicate a more complex molecular graph.

Name: HallKierAlpha
Category: regression
Value range: Real number; may be negative, zero or positive
Description: Sum of atom-size and hybridization corrections relative to a carbon reference; adjusts graph-based shape indices for different atom types and bonding states.

Name: Ipc
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: Total information content derived from the bond network's characteristic polynomial; combines structural complexity with a strong dependence on molecular size.

Name: Kappa1
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: Kier shape index of order 1, combining heavy-atom count with bond counts and atom-type corrections; describes structural shape from connectivity alone.

Name: Kappa2
Category: regression
Value range: Real number; may be negative, zero or positive
Description: Kier shape index of order 2, combining heavy-atom count with two-bond path counts and atom-type corrections; describes structural shape from connectivity alone.

Name: Kappa3
Category: regression
Value range: Real number; may be negative, zero or positive
Description: Kier shape index of order 3, combining heavy-atom count with three-bond path counts and atom-type corrections; describes structural shape from connectivity alone.

Name: Phi
Category: regression
Value range: Real number; may be negative, zero or positive
Description: Kier molecular flexibility index, calculated as Kappa1 multiplied by Kappa2 and divided by heavy-atom count; summarizes flexibility from the molecular graph.

Name: MolWt
Category: regression
Value range: 0 or greater (Da); no fixed upper limit
Description: Estimated molecular mass using average atomic weights, including hydrogens; isotope labels are respected when present in the input structure.

Name: HeavyAtomMolWt
Category: regression
Value range: 0 or greater (Da); no fixed upper limit
Description: Estimated molecular mass using average atomic weights after excluding all hydrogen contributions; isotope labels on the retained atoms are respected when supplied.

Name: ExactMolWt
Category: regression

Value range: No fixed lower or upper limit (real values, Da)
Description: Molecular mass calculated from exact isotope masses, using the most common isotope where unspecified; accounts for isotope labels and molecular charge.

Name: NumValenceElectrons
Category: count
Value range: No fixed integer bounds; negative values are possible
Description: Calculated number of valence electrons, including attached hydrogen atoms and corrections for the molecule's formal charges.

Name: NumRadicalElectrons
Category: count
Value range: 0 or greater (integers); no fixed upper limit
Description: Total number of unpaired radical electrons represented on the molecule's atoms; ordinary closed-shell structures normally have a value of zero.

Name: FractionCSP3
Category: regression
Value range: 0 to 1 inclusive
Description: Fraction of carbon atoms with sp3 hybridization, usually corresponding to tetrahedral carbon; ranges from zero to one and describes carbon saturation.

Name: HeavyAtomCount
Category: count
Value range: 0 or greater (integers); no fixed upper limit
Description: Number of atoms other than hydrogen or placeholder atoms; describes the size of the main molecular structure.

Name: NumHeteroatoms
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of atoms that are neither carbon nor hydrogen; describes the amount of other elements, such as nitrogen, oxygen, sulfur, or halogens.

Name: MaxPartialCharge
Category: regression
Value range: No fixed bounds; signed values
Description: Highest signed Gasteiger partial charge among the atoms explicitly represented in the molecular graph; reflects the most positive calculated atomic charge.

Name: MinPartialCharge
Category: regression
Value range: No fixed bounds; signed values
Description: Lowest signed Gasteiger partial charge among the atoms explicitly represented in the molecular graph; reflects the most negative calculated atomic charge.

Name: MaxAbsPartialCharge
Category: regression
Value range: 0 or greater
Description: Larger absolute value of the minimum and maximum Gasteiger atom charges; measures the greatest calculated charge magnitude among represented atoms.

Name: MinAbsPartialCharge
Category: regression
Value range: 0 or greater
Description: Smaller absolute value of the minimum and maximum Gasteiger atom charges; it compares the two signed extremes, not every atom's charge magnitude.

Name: FpDensityMorgan1
Category: regression
Value range: 0 or greater
Description: Number of distinct Morgan fingerprint features through radius 1, divided by the number of heavy atoms; summarizes local structural diversity per atom.

Name: FpDensityMorgan2
Category: regression
Value range: 0 or greater
Description: Number of distinct Morgan fingerprint features through radius 2, divided by the number of heavy atoms; summarizes local structural diversity per atom.

Name: FpDensityMorgan3
Category: regression
Value range: 0 or greater
Description: Number of distinct Morgan fingerprint features through radius 3, divided by the number of heavy atoms; summarizes local structural diversity per atom.

Name: BCUT2D_MWHI
Category: regression
Value range: 0 or greater
Description: Largest eigenvalue of a bond-network matrix weighted by atomic mass; compresses the distribution of that property across the molecule.

Name: BCUT2D_MWLOW
Category: regression
Value range: No fixed bounds; signed values
Description: Smallest eigenvalue of a bond-network matrix weighted by atomic mass; compresses the distribution of that property across the molecule.

Name: BCUT2D_CHGHI
Category: regression
Value range: No fixed bounds; signed values
Description: Largest eigenvalue of a bond-network matrix weighted by Gasteiger partial charge; compresses the distribution of that property across the molecule.

Name: BCUT2D_CHGLO
Category: regression
Value range: No fixed bounds; signed values
Description: Smallest eigenvalue of a bond-network matrix weighted by Gasteiger partial charge; compresses the distribution of that property across the molecule.

Name: BCUT2D_LOGPHI
Category: regression
Value range: No fixed bounds; signed values
Description: Largest eigenvalue of a bond-network matrix weighted by atomic contributions to logP; compresses the distribution of that property across the molecule.

Name: BCUT2D_LOGPLOW
Category: regression
Value range: No fixed bounds; signed values
Description: Smallest eigenvalue of a bond-network matrix weighted by atomic contributions to logP; compresses the distribution of that property across the molecule.

Name: BCUT2D_MRHI
Category: regression
Value range: 0 or greater

Description: Largest eigenvalue of a bond-network matrix weighted by atomic contributions to molar refractivity; compresses the distribution of that property across the molecule.

Name: BCUT2D_MRLOW
Category: regression
Value range: No fixed bounds; signed values
Description: Smallest eigenvalue of a bond-network matrix weighted by atomic contributions to molar refractivity; compresses the distribution of that property across the molecule.

Name: Chi0
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: Zero-order connectivity index that combines each atom's number of neighbors; summarizes molecular size and branching using the hydrogen-suppressed bond network.

Name: Chi0n
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: Connectivity score over individual atoms, weighting atoms by their outer-shell electron count minus attached hydrogens; describes bonding and chemical composition together.

Name: Chi0v
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: Valence connectivity score over individual atoms; weights atoms by valence-electron availability and corrects heavier elements, capturing chemical composition as well as bonding.

Name: Chi1
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: First-order connectivity index that combines the neighbor counts at both ends of every bond; summarizes branching and connectivity across atom pairs.

Name: Chi1n
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: Connectivity score over bonded atom pairs, weighting atoms by their outer-shell electron count minus attached hydrogens; describes bonding and chemical composition together.

Name: Chi1v
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: Valence connectivity score over bonded atom pairs; weights atoms by valence-electron availability and corrects heavier elements, capturing chemical composition as well as bonding.

Name: Chi2n
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: Connectivity score over two-bond paths, weighting atoms by their outer-shell electron count minus attached hydrogens; describes bonding and chemical composition together.

Name: Chi2v
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: Valence connectivity score over two-bond paths; weights atoms by valence-electron availability and corrects heavier elements, capturing chemical composition as well as bonding.

Name: Chi3n
Category: regression

Value range: Nonnegative real number; no fixed upper limit
Description: Connectivity score over three-bond paths, weighting atoms by their outer-shell electron count minus attached hydrogens; describes bonding and chemical composition together.

Name: Chi3v
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: Valence connectivity score over three-bond paths; weights atoms by valence-electron availability and corrects heavier elements, capturing chemical composition as well as bonding.

Name: Chi4n
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: Connectivity score over four-bond paths, weighting atoms by their outer-shell electron count minus attached hydrogens; describes bonding and chemical composition together.

Name: Chi4v
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: Valence connectivity score over four-bond paths; weights atoms by valence-electron availability and corrects heavier elements, capturing chemical composition as well as bonding.

Name: LabuteASA
Category: regression
Value range: Real number; may be negative, zero or positive
Description: Approximate molecular surface area estimated from atom sizes and bonding using the Labute method, including the default hydrogen contribution; no 3D coordinates are required.

Name: TPSA
Category: regression
Value range: 0 or greater (Å²); no fixed upper limit
Description: Topological polar surface area from the default nitrogen and oxygen atom fragments; estimates the molecule's polar surface without requiring 3D coordinates.

Name: PEOE_VSA1
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose Gasteiger partial charge lies in bin 1: q < -0.3.

Name: PEOE_VSA10
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose Gasteiger partial charge lies in bin 10: 0.1 <= q < 0.15.

Name: PEOE_VSA11
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose Gasteiger partial charge lies in bin 11: 0.15 <= q < 0.2.

Name: PEOE_VSA12
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose Gasteiger partial charge lies in bin 12: 0.2 <= q < 0.25.

Name: PEOE_VSA13
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose Gasteiger partial charge lies in bin 13: 0.25 <= q < 0.3.

Name: PEOE_VSA14
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose Gasteiger partial charge lies in bin 14: q >= 0.3.

Name: PEOE_VSA2
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose Gasteiger partial charge lies in bin 2: -0.3 <= q < -0.25.

Name: PEOE_VSA3
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose Gasteiger partial charge lies in bin 3: -0.25 <= q < -0.2.

Name: PEOE_VSA4
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose Gasteiger partial charge lies in bin 4: -0.2 <= q < -0.15.

Name: PEOE_VSA5
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose Gasteiger partial charge lies in bin 5: -0.15 <= q < -0.1.

Name: PEOE_VSA6
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose Gasteiger partial charge lies in bin 6: -0.1 <= q < -0.05.

Name: PEOE_VSA7
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose Gasteiger partial charge lies in bin 7: -0.05 <= q < 0.

Name: PEOE_VSA8
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose Gasteiger partial charge lies in bin 8: 0 <= q < 0.05.

Name: PEOE_VSA9
Category: regression
Value range: No fixed bounds; signed values

Description: Sum of approximate atom surface areas for atoms whose Gasteiger partial charge lies in bin 9: 0.05 <= q < 0.1.

Name: SMR_VSA1
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic molar-refractivity contribution lies in bin 1: r < 1.29.

Name: SMR_VSA10
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic molar-refractivity contribution lies in bin 10: r >= 4.

Name: SMR_VSA2
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic molar-refractivity contribution lies in bin 2: 1.29 <= r < 1.82.

Name: SMR_VSA3
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic molar-refractivity contribution lies in bin 3: 1.82 <= r < 2.24.

Name: SMR_VSA4
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic molar-refractivity contribution lies in bin 4: 2.24 <= r < 2.45.

Name: SMR_VSA5
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic molar-refractivity contribution lies in bin 5: 2.45 <= r < 2.75.

Name: SMR_VSA6
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic molar-refractivity contribution lies in bin 6: 2.75 <= r < 3.05.

Name: SMR_VSA7
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic molar-refractivity contribution lies in bin 7: 3.05 <= r < 3.63.

Name: SMR_VSA8
Category: regression
Value range: 0 only
Description: Sum of approximate atom surface areas for atoms whose atomic molar-refractivity contribution lies in bin 8: 3.63 <= r < 3.8.

Name: SMR_VSA9
Category: regression

Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic molar-refractivity contribution lies in bin 9: 3.8 <= r < 4.

Name: SlogP_VSA1
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic logP contribution lies in bin 1: p < -0.4.

Name: SlogP_VSA10
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic logP contribution lies in bin 10: 0.4 <= p < 0.5.

Name: SlogP_VSA11
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic logP contribution lies in bin 11: 0.5 <= p < 0.6.

Name: SlogP_VSA12
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic logP contribution lies in bin 12: p >= 0.6.

Name: SlogP_VSA2
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic logP contribution lies in bin 2: -0.4 <= p < -0.2.

Name: SlogP_VSA3
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic logP contribution lies in bin 3: -0.2 <= p < 0.

Name: SlogP_VSA4
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic logP contribution lies in bin 4: 0 <= p < 0.1.

Name: SlogP_VSA5
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic logP contribution lies in bin 5: 0.1 <= p < 0.15.

Name: SlogP_VSA6
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic logP contribution lies in bin 6: 0.15 <= p < 0.2.

Name: SlogP_VSA7
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic logP contribution lies in bin 7: 0.2 <= p < 0.25.

Name: SlogP_VSA8
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose atomic logP contribution lies in bin 8: 0.25 <= p < 0.3.

Name: SlogP_VSA9
Category: regression
Value range: 0 only
Description: Sum of approximate atom surface areas for atoms whose atomic logP contribution lies in bin 9: 0.3 <= p < 0.4.

Name: EState_VSA1
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose electrotopological state lies in bin 1: s < -0.39.

Name: EState_VSA10
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose electrotopological state lies in bin 10: 9.17 <= s < 15.

Name: EState_VSA11
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose electrotopological state lies in bin 11: s >= 15.

Name: EState_VSA2
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose electrotopological state lies in bin 2: -0.39 <= s < 0.29.

Name: EState_VSA3
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose electrotopological state lies in bin 3: 0.29 <= s < 0.717.

Name: EState_VSA4
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose electrotopological state lies in bin 4: 0.717 <= s < 1.165.

Name: EState_VSA5
Category: regression
Value range: No fixed bounds; signed values

Description: Sum of approximate atom surface areas for atoms whose electrotopological state lies in bin 5: 1.165 <= s < 1.54.

Name: EState_VSA6
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose electrotopological state lies in bin 6: 1.54 <= s < 1.807.

Name: EState_VSA7
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose electrotopological state lies in bin 7: 1.807 <= s < 2.05.

Name: EState_VSA8
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose electrotopological state lies in bin 8: 2.05 <= s < 4.69.

Name: EState_VSA9
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of approximate atom surface areas for atoms whose electrotopological state lies in bin 9: 4.69 <= s < 9.17.

Name: VSA_EState1
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of atom EState values for atoms whose approximate surface area lies in bin 1: a < 4.78 square angstroms.

Name: VSA_EState10
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of atom EState values for atoms whose approximate surface area lies in bin 10: a >= 11 square angstroms.

Name: VSA_EState2
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of atom EState values for atoms whose approximate surface area lies in bin 2: 4.78 <= a < 5 square angstroms.

Name: VSA_EState3
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of atom EState values for atoms whose approximate surface area lies in bin 3: 5 <= a < 5.41 square angstroms.

Name: VSA_EState4
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of atom EState values for atoms whose approximate surface area lies in bin 4: 5.41 <= a < 5.74 square angstroms.

Name: VSA_EState5
Category: regression

Value range: No fixed bounds; signed values
Description: Sum of atom EState values for atoms whose approximate surface area lies in bin 5: 5.74 <= a < 6 square angstroms.

Name: VSA_EState6
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of atom EState values for atoms whose approximate surface area lies in bin 6: 6 <= a < 6.07 square angstroms.

Name: VSA_EState7
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of atom EState values for atoms whose approximate surface area lies in bin 7: 6.07 <= a < 6.45 square angstroms.

Name: VSA_EState8
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of atom EState values for atoms whose approximate surface area lies in bin 8: 6.45 <= a < 7 square angstroms.

Name: VSA_EState9
Category: regression
Value range: No fixed bounds; signed values
Description: Sum of atom EState values for atoms whose approximate surface area lies in bin 9: 7 <= a < 11 square angstroms.

Name: NHOHCount
Category: count
Value range: 0 or greater (integers); no fixed upper limit
Description: Total number of hydrogens attached to nitrogen or oxygen atoms; an NH2 group contributes two, so this differs from counting donor sites.

Name: NOCount
Category: count
Value range: 0 or greater (integers); no fixed upper limit
Description: Total number of nitrogen and oxygen atoms; this simple Lipinski acceptor proxy includes atoms that may not actually accept hydrogen bonds.

Name: NumHAcceptors
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of atoms recognized as hydrogen-bond acceptors by RDKit's chemical rules; generally counts suitable nitrogen, oxygen, and selected other atoms.

Name: NumHDonors
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of hydrogen-bond donor sites recognized by RDKit's chemical rules; each eligible atom is counted once even when it carries multiple hydrogens.

Name: NumAliphaticCarbocycles
Category: count
Value range: 0 or greater (integers); no fixed upper limit
Description: Number of perceived rings made entirely of carbon atoms with at least one non-aromatic ring bond; counts each qualifying ring in RDKit's ring set.

Name: NumAliphaticHeterocycles
Category: count
Value range: 0 or greater (integers); no fixed upper limit
Description: Number of perceived rings containing at least one non-carbon atom with at least one non-aromatic ring bond; counts each qualifying ring in RDKit's ring set.

Name: NumAliphaticRings
Category: count
Value range: 0 or greater (integers); no fixed upper limit
Description: Number of perceived rings of any atom composition with at least one non-aromatic ring bond; counts each qualifying ring in RDKit's ring set.

Name: NumAromaticCarbocycles
Category: count
Value range: 0 or greater (integers); no fixed upper limit
Description: Number of perceived rings made entirely of carbon atoms with all ring bonds aromatic; counts each qualifying ring in RDKit's ring set.

Name: NumAromaticHeterocycles
Category: count
Value range: 0 or greater (integers); no fixed upper limit
Description: Number of perceived rings containing at least one non-carbon atom with all ring bonds aromatic; counts each qualifying ring in RDKit's ring set.

Name: NumAromaticRings
Category: count
Value range: 0 or greater (integers); no fixed upper limit
Description: Number of perceived rings of any atom composition with all ring bonds aromatic; counts each qualifying ring in RDKit's ring set.

Name: NumHeterocycles
Category: count
Value range: 0 or greater (integers); no fixed upper limit
Description: Number of perceived rings containing at least one atom other than carbon; includes aromatic, saturated, and other non-aromatic heterocycles.

Name: NumSaturatedCarbocycles
Category: count
Value range: 0 or greater (integers); no fixed upper limit
Description: Number of perceived rings made entirely of carbon atoms with only single, non-aromatic ring bonds; counts each qualifying ring in RDKit's ring set.

Name: NumSaturatedHeterocycles
Category: count
Value range: 0 or greater (integers); no fixed upper limit
Description: Number of perceived rings containing at least one non-carbon atom with only single, non-aromatic ring bonds; counts each qualifying ring in RDKit's ring set.

Name: NumSaturatedRings
Category: count
Value range: 0 or greater (integers); no fixed upper limit
Description: Number of perceived rings of any atom composition with only single, non-aromatic ring bonds; counts each qualifying ring in RDKit's ring set.

Name: RingCount
Category: count
Value range: 0 or greater (integers); no fixed upper limit

Description: Number of rings in RDKit's perceived ring set; summarizes cyclic structure and follows RDKit's ring perception rather than counting every possible cycle.

Name: NumAmideBonds
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of carbonyl-carbon-to-nitrogen amide bonds recognized by RDKit; includes these linkages in both open-chain structures and cyclic amides.

Name: NumRotatableBonds
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of bonds considered rotatable under RDKit's default definition; estimates flexibility while excluding ring bonds and restricted linkages such as amides.

Name: NumAtomStereoCenters
Category: count
Value range: 0 or greater (integers); no fixed upper limit
Description: Total number of atomic stereocenters recognized by RDKit, including centers with specified stereochemistry and potential centers whose configuration is not specified.

Name: NumBridgeheadAtoms
Category: count
Value range: 0 or greater (integers); no fixed upper limit
Description: Number of atoms at the ends of ring bridges, identified from rings sharing multiple bonds; measures junctions in bridged ring systems.

Name: NumSpiroAtoms
Category: count
Value range: 0 or greater (integers); no fixed upper limit
Description: Number of spiro atoms, where two rings meet at exactly one shared atom; counts the junction atoms rather than the rings.

Name: NumUnspecifiedAtomStereoCenters
Category: count
Value range: 0 or greater (integers); no fixed upper limit
Description: Number of recognized atomic stereocenters whose configuration is left unspecified in the input; excludes stereocenters with an assigned configuration.

Name: fr_Al_COO
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of carboxylic acid or carboxylate groups whose carbonyl carbon is attached to a non-aromatic carbon; counts both neutral and deprotonated forms.

Name: fr_Al_OH
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of hydroxyl groups attached to non-aromatic carbon outside a carbonyl group; counts alcohol-like OH groups while excluding ordinary carboxylic acid hydroxyls.

Name: fr_Al_OH_noTert
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of non-aromatic carbon atoms bearing an OH group, excluding carboxylic acids and tertiary alcohols; a carbon bearing multiple OH groups is counted once.

Name: fr_ArN
Category: count

Value range: 0 to 1,000 (integers)
Description: Number of selected nitrogen-containing groups attached to aromatic systems, including amino, hydrazino, hydrazide, and amidine patterns; uses RDKit's specific built-in structural rules.

Name: fr_Ar_COO
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of carboxylic acid or carboxylate groups whose carbonyl carbon is attached directly to an aromatic carbon; includes both neutral and deprotonated forms.

Name: fr_Ar_N
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of nitrogen atoms that belong to aromatic systems, including pyridine-like nitrogen and hydrogen-bearing aromatic nitrogen; counts atoms rather than whole rings.

Name: fr_Ar_NH
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of aromatic nitrogen atoms carrying one hydrogen, represented by the [nH] pattern; gives the same count as fr_Nhpyrrole in this RDKit release.

Name: fr_Ar_OH
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of hydroxyl groups attached directly to an aromatic carbon atom; includes phenol-like groups on aromatic systems without requiring a benzene-only ring.

Name: fr_COO
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of carboxylic acid or carboxylate groups with a carbon substituent on the carbonyl carbon; includes neutral acids and their singly deprotonated forms.

Name: fr_COO2
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of acid or carboxylate-like carbonyl groups without requiring a carbon substituent on the carbonyl carbon; this pattern is broader than fr_COO.

Name: fr_C_O
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of carbonyl groups containing a carbon-oxygen double bond; includes matching groups in aldehydes, ketones, carboxylic acids, esters, amides, and related structures.

Name: fr_C_O_noCOO
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of carbonyl groups excluding carbonyl carbons bonded to an OH group; carboxylate anions can still match, as can esters and amides.

Name: fr_C_S
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of thiocarbonyl groups containing a carbon-sulfur double bond; includes matching groups in thioketones, thioamides, thioureas, and related sulfur-containing structures.

Name: fr_HOCCN
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of hydroxyethylamine-like arrangements with OH-C-C-N connectivity where nitrogen is in a ring or attached to a tert-butyl carbon group.

Name: fr_Imine
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of imine-like groups with a non-aromatic nitrogen double-bonded to carbon and single-bonded to another carbon; excludes aromatic ring nitrogens.

Name: fr_NH0
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of nitrogen atoms with no attached hydrogens, including aromatic nitrogen; although historically labeled tertiary amines, this broad pattern matches many other nitrogen groups.

Name: fr_NH1
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of nitrogen atoms with one attached hydrogen, including aromatic forms; a hydrogen-count feature that can match secondary amines, amides, and other nitrogen groups.

Name: fr_NH2
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of nitrogen atoms bearing two hydrogens, including amide nitrogens and allowed aromatic forms; this pattern is broader than a primary-amine-only count.

Name: fr_N_O
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of hydroxylamine-like nitrogen-oxygen single-bond patterns with nitrogen also attached to a non-aromatic carbon; excludes nitrogen atoms that are double-bonded to oxygen.

Name: fr_Ndealkylation1
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of nitrogen sites with selected methyl or ethyl substituents and a second carbon chain matching RDKit's N-dealkylation pattern; indicates structural sites, not measured metabolism.

Name: fr_Ndealkylation2
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of N-substituted saturated nitrogen rings of four to eight members matching RDKit's pattern; excludes extra ring heteroatoms and certain bridged nitrogen arrangements.

Name: fr_Nhpyrrole
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of aromatic nitrogen atoms bearing one hydrogen, as in pyrrole; uses the same [nH] pattern and gives the same count as fr_Ar_NH.

Name: fr_SH
Category: count
Value range: 0 to 1,000 (integers)

Description: Number of sulfur atoms bearing one hydrogen, corresponding to thiol-like SH groups; counts matching atoms rather than the molecule's total sulfur content.

Name: fr_aldehyde
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of aldehyde carbonyl groups whose carbonyl carbon carries one hydrogen and a carbon substituent; excludes formaldehyde because it lacks that carbon substituent.

Name: fr_alkyl_carbamate
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of carbamate arrangements with carbon-NH-C(=O)-O-carbon connectivity; counts the specific N-H-containing alkyl carbamate pattern used by RDKit.

Name: fr_alkyl_halide
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of bonds from a tetrahedral carbon atom to fluorine, chlorine, bromine, or iodine; counts alkyl-halide attachments rather than all halogen atoms.

Name: fr_allylic_oxid
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of selected carbon-carbon-double-bond-adjacent patterns associated with allylic oxidation sites; excludes specified nearby heteroatoms and a steroid dienone pattern in RDKit's rules.

Name: fr_amide
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of carbonyl-carbon-to-nitrogen amide linkages, including open-chain and cyclic examples; counts structural matches rather than checking whether the molecule is an amide overall.

Name: fr_amidine
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of amidine-like carbon centers bonded to one nitrogen by a double bond and another by a single bond; excludes guanidine-type centers.

Name: fr_aniline
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of non-aromatic, three-coordinate nitrogen groups attached directly to aromatic carbon, excluding nitrogen with a double bond; represents aniline-like aromatic amine linkages.

Name: fr_aryl_methyl
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of selected methyl or short alkyl side-chain patterns on aromatic systems associated with hydroxylation sites; the RDKit pattern is broader than a simple aryl-methyl count.

Name: fr_azide
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of azide groups represented by the supported charge-separated three-nitrogen patterns; counts the recognized substructures rather than measuring their chemical reactivity.

Name: fr_azo
Category: count

Value range: 0 to 1,000 (integers)
Description: Number of azo linkages with carbon-N=N-carbon connectivity; requires carbon attachments on both ends and therefore excludes azide groups from this count.

Name: fr_barbitur
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of occurrences of the barbiturate ring pattern, a six-membered ring with two nitrogen atoms and three carbonyl groups in the specified arrangement.

Name: fr_benzene
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of six-carbon aromatic ring patterns matching benzene; includes matching rings within larger fused aromatic systems and counts each unique ring atom set.

Name: fr_benzodiazepine
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of occurrences of RDKit's specific benzodiazepine ring pattern, with a benzene fused to a seven-membered nitrogen-containing ring and no additional fused rings.

Name: fr_bicyclic
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of bonds between two atoms that each belong to exactly two perceived rings; this is a ring-junction pattern count, not a count of bicyclic molecules.

Name: fr_diazo
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of occurrences of the positively charged nitrogen triple-bonded to nitrogen pattern, [N+]#N; this exact RDKit pattern may include diazonium-like representations.

Name: fr_dihydropyridine
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of six-membered nitrogen-containing rings matching RDKit's listed dihydropyridine patterns; recognizes specified partially unsaturated arrangements rather than every possible nitrogen ring tautomer.

Name: fr_epoxide
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of three-membered rings containing one oxygen and two carbon atoms with single bonds; counts the strained cyclic ether pattern known as an epoxide.

Name: fr_ester
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of ester groups with carbon-C(=O)-O-carbon connectivity; requires carbon attachments on both sides of the ester linkage in RDKit's pattern.

Name: fr_ether
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of oxygen atoms with two carbon neighbors, joined by single bonds; includes aromatic ethers and can also match ester-like oxygen environments.

Name: fr_furan
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of five-membered aromatic rings containing one oxygen and four carbon atoms; includes furan patterns embedded within larger fused ring systems.

Name: fr_guanido
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of guanidine-like carbon centers joined to three nitrogen atoms, one through a double bond and two through single bonds, as represented in the input.

Name: fr_halogen
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of fluorine, chlorine, bromine, and iodine atoms; counts these four halogens whether attached to carbon or present in another chemical environment.

Name: fr_hdrzine
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of single bonds between two three-coordinate nitrogen atoms, representing hydrazine-like groups; can include substituted, cyclic, and carbonyl-adjacent hydrazine arrangements.

Name: fr_hdrzone
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of hydrazone-like carbon=N-N arrangements where the terminal nitrogen is three-coordinate; recognizes both open-chain and cyclic matching structures.

Name: fr_imidazole
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of five-membered aromatic rings with two nitrogen atoms separated by one carbon, matching imidazole; includes these rings within larger fused systems.

Name: fr_imide
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of nitrogen atoms bonded to two carbonyl carbons, matching an imide arrangement; includes cyclic and open-chain occurrences of the structural pattern.

Name: fr_isocyan
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of isocyanate groups with nitrogen-carbon-oxygen connectivity and two successive double bonds, N=C=O; counts each matching group in the molecular structure.

Name: fr_isothiocyan
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of isothiocyanate groups with nitrogen-carbon-sulfur connectivity and two successive double bonds, N=C=S; counts each matching group in the molecular structure.

Name: fr_ketone
Category: count
Value range: 0 to 1,000 (integers)

Description: Number of ketone groups in which a carbonyl carbon is bonded to two other carbon atoms; counts both aliphatic and aromatic carbon substituents.

Name: fr_ketone_Topliss
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of ketones matching RDKit's selected Topliss pattern, requiring at least one non-aromatic carbon substituent and excluding a specified directly conjugated alkene arrangement.

Name: fr_lactam
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of four-membered cyclic amide rings matching the beta-lactam pattern; despite its short name, this descriptor does not count all possible lactam ring sizes.

Name: fr_lactone
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of cyclic ester patterns where the carbonyl carbon, ester oxygen, and neighboring carbon each belong to one perceived ring; counts matching lactone-like arrangements.

Name: fr_methoxy
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of O-CH3 groups whose oxygen is also attached to another carbon; includes methyl ethers, aromatic methoxy groups, and matching methyl ester groups.

Name: fr_morpholine
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of six-membered saturated rings containing one oxygen and one nitrogen at opposite positions; counts the ring pattern characteristic of morpholine.

Name: fr_nitrile
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of carbon-nitrogen triple bonds with the nitrile atom-connectivity pattern, C#N; counts nitrile groups rather than all nitrogen-containing multiple bonds.

Name: fr_nitro
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of nitro groups attached to a non-oxygen atom, recognizing the supported neutral or charge-separated representations; counts the matching nitrogen-dioxide-containing groups.

Name: fr_nitro_arom
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of nitro substituents attached to a six-carbon aromatic ring; counts the nitrobenzene-like attachment sites recognized by RDKit's built-in pattern.

Name: fr_nitro_arom_nonortho
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of nitrobenzene-like attachment sites without an additional ortho substituent according to RDKit's pattern; focuses on the local substitution arrangement around the nitro group.

Name: fr_nitroso
Category: count

Value range: 0 to 1,000 (integers)
Description: Number of nitrogen-oxygen double-bond groups where that nitrogen is not also single-bonded to oxygen; counts nitroso-like groups while excluding ordinary nitro groups.

Name: fr_oxazole
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of five-membered aromatic rings containing one oxygen and one nitrogen in the oxazole arrangement; includes matching rings within larger fused systems.

Name: fr_oxime
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of carbon=N-O patterns with the specified atom coordination; covers oxime-like groups and matching O-substituted derivatives as represented in the molecule.

Name: fr_para_hydroxylation
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of selected unsubstituted aromatic positions opposite oxygen- or nitrogen-containing substituents; counts RDKit pattern matches associated with potential para-hydroxylation sites.

Name: fr_phenol
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of hydroxyl groups attached to a six-carbon aromatic ring, matching the phenol pattern; narrower than the general aromatic-hydroxyl descriptor fr_Ar_OH.

Name: fr_phenol_noOrthoHbond
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of phenol-like sites excluding specified neighboring hydroxymethyl, carboxyl, and primary-amide substituents that can support an internal hydrogen bond in RDKit's pattern.

Name: fr_phos_acid
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of phosphoric-acid-like phosphorus centers with hydroxyl, negatively charged oxygen, or phosphate-bridge substituents; recognizes supported neutral and charge-separated phosphate representations.

Name: fr_phos_ester
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of phosphate ester centers containing at least one phosphorus-O-carbon attachment; recognizes the supported neutral, charged, and selected phosphate-bridge arrangements in RDKit's pattern.

Name: fr_piperdine
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of six-membered saturated rings with one nitrogen and five carbon atoms, corresponding to piperidine; preserves RDKit's original descriptor spelling in the name.

Name: fr_piperzine
Category: count
Value range: 0 to 1,000 (integers)

Description: Number of six-membered saturated rings with two nitrogen atoms at opposite positions, corresponding to piperazine; preserves RDKit's original descriptor spelling in the name.

Name: fr_priamide
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of primary amide groups with a carbonyl carbon bonded to NH2; counts these groups separately from amides whose nitrogen carries fewer hydrogens.

Name: fr_prisulfonamd
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of primary sulfonamide groups with NH2 attached to a carbon-substituted sulfonyl sulfur; recognizes the supported neutral or charge-separated sulfur-oxygen representations.

Name: fr_pyridine
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of six-membered aromatic rings containing one nitrogen and five carbon atoms; includes pyridine patterns within larger fused aromatic ring systems.

Name: fr_quatN
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of nitrogen centers matching RDKit's four-coordinate positively charged or specified four-coordinate double-bonded pattern; includes the common quaternary-ammonium structural arrangement.

Name: fr_sulfide
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of two-coordinate sulfur bonded to two carbon atoms, with at least one non-aromatic carbon; counts selected thioethers while excluding oxidized sulfur groups.

Name: fr_sulfonamd
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of sulfonamide N-S(oxygen)2-carbon patterns, including supported neutral and charge-separated sulfur-oxygen representations; counts these nitrogen-to-sulfonyl linkages throughout the molecule.

Name: fr_sulfone
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of sulfone-like sulfur centers attached to two oxygen atoms and two carbons; includes supported charged representations and excludes sulfonamide or sulfonyl-halide arrangements.

Name: fr_term_acetylene
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of terminal carbon-carbon triple-bond groups where one triple-bonded carbon bears a hydrogen, corresponding to the terminal alkyne pattern C#CH.

Name: fr_tetrazole
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of five-membered aromatic rings containing four nitrogen atoms and one carbon atom; counts rings matching the tetrazole pattern, including supported substitutions.

Name: fr_thiazole
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of five-membered aromatic rings containing sulfur and nitrogen in the thiazole arrangement; includes matching rings within larger fused aromatic systems.

Name: fr_thiocyan
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of thiocyanate-like sulfur-carbon-nitrogen arrangements with an S-C single bond and C#N triple bond; distinct from the N=C=S isothiocyanate pattern.

Name: fr_thiophene
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of five-membered aromatic rings containing one sulfur and four carbon atoms; includes thiophene patterns embedded within larger fused aromatic ring systems.

Name: fr_unbrch_alkane
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of four-carbon non-ring chain segments whose internal atoms have two neighbors and ends have one or two; longer unbranched chains can contribute overlapping matches.

Name: fr_urea
Category: count
Value range: 0 to 1,000 (integers)
Description: Number of carbonyl carbons bonded to two nitrogen atoms, matching the urea arrangement; includes cyclic ureas and other molecules containing this structural unit.

Name: PMI1
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: Smallest principal moment of inertia of a chosen 3D conformer; measures how mass is distributed around the axis with least rotational inertia.

Name: PMI2
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: Middle principal moment of inertia of a chosen 3D conformer; measures mass distribution around the intermediate principal axis of the molecule.

Name: PMI3
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: Largest principal moment of inertia of a chosen 3D conformer; measures mass distribution around the axis with greatest rotational inertia.

Name: NPR1
Category: regression
Value range: 0 to 1 inclusive
Description: Ratio of the smallest to largest principal moment of inertia, PMI1/PMI3; helps distinguish rod-like, disk-like, and more spherical 3D molecular shapes.

Name: NPR2
Category: regression
Value range: 0 to 1 inclusive

Description: Ratio of the middle to largest principal moment of inertia, PMI2/PMI3; complements NPR1 when comparing the overall shape of 3D conformers.

Name: RadiusOfGyration
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: Root-mean-square distance of atoms from the conformer's center of mass, using atomic masses by default; describes how widely the structure is spread out.

Name: InertialShapeFactor
Category: regression
Value range: 0 to 10,000 inclusive
Description: Shape index calculated as PMI2 divided by the product of PMI1 and PMI3; combines the principal moments to describe a conformer's mass distribution.

Name: Eccentricity
Category: regression
Value range: 0 to 1 inclusive
Description: Measure of unequal mass distribution along a conformer's principal axes, calculated from the smallest and largest moments of inertia; zero corresponds to equal moments.

Name: Asphericity
Category: regression
Value range: 0 to 1 inclusive
Description: Measure of departure from a spherical distribution of mass, calculated from the gyration tensor; zero indicates equal spatial spread in all directions.

Name: SpherocityIndex
Category: regression
Value range: 0 to 1 inclusive
Description: Geometric sphericity score using the unweighted gyration tensor; approaches zero for planar or linear structures and one for equal spread in three directions.

Name: PBF
Category: regression
Value range: Nonnegative real number; no fixed upper limit
Description: Mean perpendicular distance of the conformer's atoms from their best-fitting plane; zero indicates a planar structure, while larger values indicate greater three-dimensionality.